Tutorial 19/02

1. Consider the following two samples. The first was generated from a $\mathcal{N}(50, 64)$ distribution while the second was generated from a $\mathcal{N}(50, 144)$ distribution. Thus the ratio of scales is 2.0.
   Sample from N(50, 64) distribution.

$$43.72, 58.06, 55.57, 57.49, 64.16, 43.49, 49.94, 49.94$$

$$52.58, 49.40, 38.93, 42.65, 42.91, 46.59, 47.88$$

   Sample from N(50, 256) distribution.

$$77.10, 42.94, 42.32, 53.51, 66.79, 18.26, 38.72, 29.04,$$

$$33.54, 61.19, 32.87, 60.98, 59.02, 60.68, 45.11, 43.52, 35.44, 34.31$$

   (a) Compare the spreads visually using boxplots.

   (b) Obtain the analysis based on the Fligner–Killeen procedure to test for difference in scales for this data. Was the confidence interval successful in trapping the true ratio of scales given by $\eta = 2.0$?

2. The following code generates a sample from this double exponential distribution.

```
rlaplace<-function(n){
x<-rexp(n)
ind<-sample(c(-1,1),n,replace=TRUE)
ind*x
}
```

   Use sample sizes of $n_1 = 30$ and $n_2 = 35$, and sample under H_0; i.e., draw BOTH samples using the R function rlaplace.

   (a) Carry out the F test for equality of variances and record the p-value.

   (b) Carry out the Fligner-Killen test for equality of variances and record the p-value.

   Repeat the above procedure 1000 times and compare the performance of the parametric and non-parametric procedures.

3. This is regarding Behavioral Neuroscience (Feb. 2004) study of the drug scopolamine's effects on memory for word-pair association. Three groups was used: Group 1 subjects were injected with scopolamine, group 2 subjects were injected with a placebo, and group 3 subjects were not given any drug. The response variable was number of word pairs recalled.The data on all 28 subjects are reproduced below.
   Group 1 (Scopolamine): 5 8 8 6 6 6 6 8 6 4 5 6
   Group 2 (Placebo): 8 10 12 10 9 7 9 10
   Group 3 (No drug): 8 9 11 12 11 10 12 12

(a) Rank the data for all 28 observations from smallest to largest.

(b) Sum the ranks of the observations from group 1.

(c) Sum the ranks of the observations from group 2.

(d) Sum the ranks of the observations from group 3.

(e) Use the rank sums to compute the Kruskal–Wallis H statistic.

(f) Carry out the Kruskal–Wallis nonparametric test (at $\alpha = 5\%$) to compare the distributions of number of word pairs recalled for the three groups.